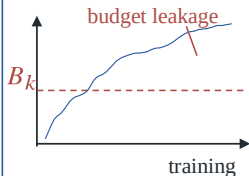


Penalty view

Costs weight the actor

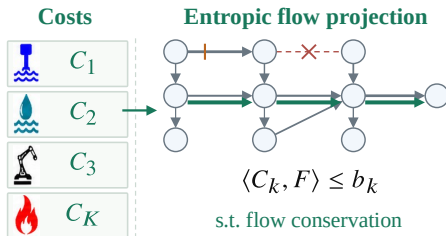
$$A_r - \sum_k \lambda_k A_{c_k}$$



Fixed price can leave a budget violated.
(schematic)

SafeOT

One flow prices all budgets



gray: observed · green: target

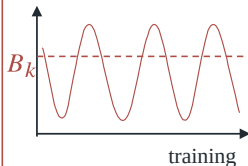
× hard constraint: excluded edge

Prices before violations

Constraint view

Budgets bound costs

$$J_{c_k}(\pi) \leq B_k$$



Lagrangian feedback can lag and oscillate.
(schematic)

$\lambda_k^* \rightarrow \text{clip} + \text{EMA} \rightarrow$

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

← observed-cost feedback

→ priced actor

PPO/TRPO → π_{t+1}

One price rule

Fixed penalty

$c = 0, \beta_k$ fixed

exact settings

Lagrangian

$c = 0, \beta_k$ adaptive

SafeOT

finite ε

LP

$\varepsilon \rightarrow 0$

conditional limits

Active-set

$\varepsilon \rightarrow \infty$, saturated clip